Topic: Singular points and Frobenius' Theorem

Question: Find and classify any singular points of the differential equation.

$$(x^2 - 9)y'' + (x + 3)y' + 2y = 0$$

Answer choices:

A $x_0 = 3$ and $x_0 = -3$ are regular singular points

B $x_0 = 3$ and $x_0 = -3$ are irregular singular points

C $x_0 = -3$ is a regular singular point; $x_0 = 3$ is an irregular singular point

D $x_0 = 3$ is a regular singular point; $x_0 = -3$ is an irregular singular point

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Solution: A

The differential equation

$$(x^2 - 9)y'' + (x + 3)y' + 2y = 0$$

has singular points at

$$x^2 - 9 = 0$$

$$x^2 = 9$$

$$x = \pm 3$$

We'll find $Q(x)$ and $R(x)$ and get

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)}$$

$$Q(x) = (x - x_0)\frac{x + 3}{x^2 - 9}$$

$$Q(x) = (x - x_0)\frac{x + 3}{(x + 3)(x - 3)}$$

$$Q(x) = (x - x_0)\frac{1}{x - 3}$$

and

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)}$$

$$R(x) = (x - x_0)^2\frac{2}{x^2 - 9}$$

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$$R(x) = (x - x_0)^2 \frac{2}{(x+3)(x-3)}$$

At $x_0 = 3$, we find

$$Q(x) = (x-3)\frac{1}{x-3}$$

$$Q(x) = 1$$

and

$$R(x) = (x-3)^2 \frac{2}{(x+3)(x-3)}$$

$$R(x) = \frac{2(x-3)}{x+3}$$

Both $Q(x)$ and $R(x)$ are analytic at $x = 3$ because

$$Q(3) = 1$$

$$R(3) = \frac{2(3-3)}{3+3} = 0$$

And at $x_0 = -3$, we find

$$Q(x) = (x-(-3))\frac{1}{x-3}$$

$$Q(x) = \frac{x+3}{x-3}$$

and

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$$R(x) = (x-(-3))^2 \frac{2}{(x+3)(x-3)}$$

$$R(x) = \frac{2(x+3)}{x-3}$$

Both $Q(x)$ and $R(x)$ are analytic at $x = -3$ because

$$Q(-3) = \frac{-3+3}{-3-3} = 0$$

$$R(-3) = \frac{2(-3+3)}{-3-3} = 0$$

So both $x_0 = 3$ and $x_0 = -3$ are regular singular points.

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Topic: Singular points and Frobenius' Theorem

Question: Use the method of Frobenius to find the indicial equation and recurrence relation of the differential equation.

$$x^2y'' + \left(\frac{5}{3}x + x^2\right)y' - \frac{1}{3}y = 0$$

Answer choices:

A $r^2 + \frac{2}{3}r - \frac{1}{3} = 0$ and $c_k = -\frac{3(k+r-1)c_{k-1}}{3k^2+2k+6kr+2r+3r^2-1}$

B $r^2 - \frac{2}{3}r + \frac{1}{3} = 0$ and $c_k = \frac{3(k+r-1)c_{k-1}}{3k^2+2k+6kr+2r+3r^2-1}$

C $r - \frac{1}{3} = 0$ and $c_k = -\frac{3(k+r-1)c_{k-1}}{3k^2+2k+6kr+2r+3r^2-1}$

D $r + \frac{1}{3} = 0$ and $c_k = \frac{3(k+r-1)c_{k-1}}{3k^2+2k+6kr+2r+3r^2-1}$

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Solution: A

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = x^2$$

$$q(x) = \frac{5}{3}x + x^2$$

$$r(x) = -\frac{1}{3}$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{\frac{5}{3}x + x^2}{x^2} = \frac{5}{3} + x$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{\left(-\frac{1}{3}\right)}{x^2} = -\frac{1}{3}$$

We can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point. Now we'll use

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n + r)(n + r - 1)c_n x^{n+r-2}$$

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to make substitutions into the differential equation.

$$x^2y'' + \left(\frac{5}{3}x + x^2\right)y' - \frac{1}{3}y = 0$$

$$x^2\sum_{n=0}^{\infty}(n+r)(n+r-1)c_nx^{n+r-2}$$

$$+\left(\frac{5}{3}x + x^2\right)\sum_{n=0}^{\infty}(n+r)c_nx^{n+r-1} - \frac{1}{3}\sum_{n=0}^{\infty}c_nx^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(n+r)(n+r-1)c_nx^{n+r} + \frac{5}{3}x\sum_{n=0}^{\infty}(n+r)c_nx^{n+r-1}$$

$$+x^2\sum_{n=0}^{\infty}(n+r)c_nx^{n+r-1} - \sum_{n=0}^{\infty}\frac{1}{3}c_nx^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(n+r)(n+r-1)c_nx^{n+r} + \sum_{n=0}^{\infty}\frac{5}{3}(n+r)c_nx^{n+r}$$

$$+\sum_{n=0}^{\infty}(n+r)c_nx^{n+r+1} - \sum_{n=0}^{\infty}\frac{1}{3}c_nx^{n+r} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty}\left[(n+r)(n+r-1)c_n + \frac{5}{3}(n+r)c_n - \frac{1}{3}c_n\right]x^{n+r}$$

$$+\sum_{n=0}^{\infty}(n+r)c_nx^{n+r+1} = 0$$

Factor x^r out of the left side of the equation.

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$$x^r\left[\sum_{n=0}^{\infty}\left((n+r)(n+r-1)c_n+\frac{5}{3}(n+r)c_n-\frac{1}{3}c_n\right)x^n+\sum_{n=0}^{\infty}(n+r)c_nx^{n+1}\right]=0$$

Now we need to make sure these two series are in phase. The first series starts with the x^0 term, while the second series starts with the x^1 term. So we'll pull the x^0 term out of the first series to put them in phase.

$$x^r\left[\left(r^2+\frac{2}{3}r-\frac{1}{3}\right)c_0+\sum_{n=1}^{\infty}\left((n+r)(n+r-1)c_n+\frac{5}{3}(n+r)c_n-\frac{1}{3}c_n\right)x^n\right.$$

$$\left.+\sum_{n=0}^{\infty}(n+r)c_nx^{n+1}\right]=0$$

Now that the series are in phase, we'll substitute $k=n$ into the first series, and $k=n+1$ and $n=k-1$ into the second series.

$$x^r\left[\left(r^2+\frac{2}{3}r-\frac{1}{3}\right)c_0+\sum_{k=1}^{\infty}\left((k+r)(k+r-1)c_k+\frac{5}{3}(k+r)c_k-\frac{1}{3}c_k\right)x^k\right.$$

$$\left.+\sum_{k=1}^{\infty}(k+r-1)c_{k-1}x^k\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[\left(r^2+\frac{2}{3}r-\frac{1}{3}\right)c_0\right.$$

$$\left.+\sum_{k=1}^{\infty}\left((k+r)(k+r-1)c_k+\frac{5}{3}(k+r)c_k-\frac{1}{3}c_k+(k+r-1)c_{k-1}\right)x^k\right]=0$$

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This equation gives

$$\left(r^2+\frac{2}{3}r-\frac{1}{3}\right)c_0=0 \qquad k=0$$

$$(k+r)(k+r-1)c_k+\frac{5}{3}(k+r)c_k-\frac{1}{3}c_k+(k+r-1)c_{k-1}=0 \qquad k=1,\,2,\,3,\,\ldots$$

So the indicial equation and recurrence relation are

$$r^2+\frac{2}{3}r-\frac{1}{3}=0$$

$$c_k=-\frac{3(k+r-1)c_{k-1}}{3k^2+2k+6kr+2r+3r^2-1}$$

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Topic: Singular points and Frobenius' Theorem

Question: Use the method of Frobenius to find the general solution to the differential equation about $x_0 = 0$.

$$4xy'' + \frac{1}{2}y' + y = 0$$

Answer choices:

A $$y(x) = C_1\left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \ldots\right) + C_2x^{\frac{7}{8}}\left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \ldots\right)$$

B $$y(x) = C_1x\left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \ldots\right) + C_2x^{\frac{7}{8}}\left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \ldots\right)$$

C $$y(x) = C_1\left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \ldots\right) + C_2x^{\frac{7}{8}}\left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \ldots\right)$$

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D $$y(x) = C_1 x\left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \ldots\right)$$

$$+C_2 x^{\frac{7}{8}}\left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \ldots\right)$$

Solution: C

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 4x$$

$$q(x) = \frac{1}{2}$$

$$r(x) = 1$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{\frac{1}{2}}{4x} = \frac{1}{8}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{1}{4x} = \frac{1}{4}x$$

We can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point. Now we'll use

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$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$4xy'' + \frac{1}{2}y' + y = 0$$

$$4x\sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} + \frac{1}{2}\sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 4(n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty} \frac{1}{2}(n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty} \left(4(n+r)(n+r-1)c_n + \frac{1}{2}(n+r)c_n\right) x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

Factor x^r out of the left side of the equation.

$$x^r \left[\sum_{n=0}^{\infty} \left(4(n+r)(n+r-1)c_n + \frac{1}{2}(n+r)c_n\right) x^{n-1} + \sum_{n=0}^{\infty} c_n x^n\right] = 0$$

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Now we need to make sure these two series are in phase. The first series starts with the x^{-1} term, while the second series starts with the x^0 term. So we'll pull the x^{-1} term out of the first series to put them in phase.

$$x^r\left[r\left(4r-\frac{7}{2}\right)c_0x^{-1}+\sum_{n=1}^{\infty}\left(4(n+r)(n+r-1)c_n+\frac{1}{2}(n+r)c_n\right)x^{n-1}\right.$$

$$\left.+\sum_{n=0}^{\infty}c_nx^n\right]=0$$

Now that the series are in phase, we'll substitute $k=n-1$ and $n=k+1$ into the first series, and $k=n$ into the second series.

$$x^r\left[r\left(4r-\frac{7}{2}\right)c_0x^{-1}+\sum_{k=0}^{\infty}\left(4(k+r+1)(k+r)c_{k+1}+\frac{1}{2}(k+r+1)c_{k+1}\right)x^k\right.$$

$$\left.+\sum_{k=0}^{\infty}c_kx^k\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[r\left(4r-\frac{7}{2}\right)c_0x^{-1}\right.$$

$$\left.+\sum_{k=0}^{\infty}\left(4(k+r+1)(k+r)c_{k+1}+\frac{1}{2}(k+r+1)c_{k+1}+c_k\right)x^k\right]=0$$

$$x^r\left[r\left(4r-\frac{7}{2}\right)c_0x^{-1}\right.$$

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$$+\sum_{k=0}^{\infty}\left(\left(4(k+r+1)(k+r)+\frac{1}{2}(k+r+1)\right)c_{k+1}+c_k\right)x^k\Bigg]=0$$

This equation gives

$$r\left(4r-\frac{7}{2}\right)c_0=0$$

$$\left(4(k+r+1)(k+r)+\frac{1}{2}(k+r+1)\right)c_{k+1}+c_k=0 \qquad k=0,\ 1,\ 2,\ \ldots$$

or

$$r\left(4r-\frac{7}{2}\right)=0$$

$$c_{k+1}=-\frac{c_k}{(k+r+1)\left(4(k+r)+\frac{1}{2}\right)} \qquad k=0,\ 1,\ 2,\ \ldots$$

The indicial equation gives us $r_1=0$ and $r_2=7/8$. Substituting these indicial roots into the recurrence relation, we get

	For $r_1=0$	For $r_2=7/8$
	$c_{k+1}=-\dfrac{2c_k}{(k+1)(8k+1)}$	$c_{k+1}=-\dfrac{2c_k}{(8k+15)(k+1)}$
$k=0$	$c_1=-2c_0$	$c_1=-\dfrac{2c_0}{15}$
$k=1$	$c_2=\dfrac{2c_0}{9}$	$c_2=\dfrac{2c_0}{345}$

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$k = 2$	$c_3 = -\dfrac{4c_0}{459}$	$c_3 = -\dfrac{4c_0}{32{,}085}$
...	...	...

Forming these coefficients into series around $x_0 = 0$ gives,

$$c_0(x - x_0)^0 + c_1(x - x_0)^1 + c_2(x - x_0)^2 + c_3(x - x_0)^3 + \dots$$

$$c_0(x - x_0)^0 - 2c_0(x - x_0)^1 + \frac{2c_0}{9}(x - x_0)^2 - \frac{4c_0}{459}(x - x_0)^3 + \dots$$

$$c_0\left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \dots\right)$$

and

$$c_0(x - x_0)^0 + c_1(x - x_0)^1 + c_2(x - x_0)^2 + c_3(x - x_0)^3 + \dots$$

$$c_0(x - x_0)^0 - \frac{2c_0}{15}(x - x_0)^1 + \frac{2c_0}{345}(x - x_0)^2 - \frac{4c_0}{32{,}085}(x - x_0)^3 + \dots$$

$$c_0\left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \dots\right)$$

so the series solutions are

$$y_1(x) = c_0x^0\left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \dots\right)$$

$$y_2(x) = c_0x^{\frac{7}{8}}\left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \dots\right)$$

and the general solution on the interval $(0,\infty)$ is

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$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$y(x) = C_1 c_0 \left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \dots\right)$$

$$+ C_2 c_0 x^{\frac{7}{8}} \left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \dots\right)$$

But $C_1 c_0$ and $C_2 c_0$ are constants, so we can simplify them and write the general solution as

$$y(x) = C_1 \left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \dots\right)$$

$$+ C_2 x^{\frac{7}{8}} \left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32{,}085}x^3 + \dots\right)$$

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